**DO NOW — RECALL**

# Lesson 075 · Big O: Quadratic, Exponential and Growth

Name: _____ Class: _____ Date: _____

**1. How does O(n) differ from O(1)?**

**2. How many halvings 32→1?**

**3. Which algorithm naturally explores include/exclude branches?**

**4. A rectangular 4×3 zero-indexed array holds how many elements? Give its last valid index pair.**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

One grows with n; the other does not.

### 2. Model answer

Five.

### 3. Model answer

Subset backtracking or exhaustive enumeration.

### 4. Model answer

12 elements; last index (3,2).

### By the end, you can…

• Two nested n loops perform n×n operations, producing $O(n^2)$ time when each inner operation is constant.

• A binary decision at each of n positions can create $2^n$ possibilities. This exponential growth quickly outruns polynomial methods.

• Big O ignores constants and lower-order terms; compare classes by input growth while acknowledging memory, workload and implementation detail.

### Vocabulary — know these terms

quadratic — Growth proportional to $n^2$.

exponential — Growth like $2^n$.

pruning — Avoid exploring impossible branches.

scalability — Handling larger inputs feasibly.

NEW LEARNING

# Read, annotate and explain

### Observe

Two nested n loops perform n×n operations, producing $O(n^2)$ time when each inner operation is constant.

### Explain

A binary decision at each of n positions can create $2^n$ possibilities. This exponential growth quickly outruns polynomial methods.

### Evaluate

Big O ignores constants and lower-order terms; compare classes by input growth while acknowledging memory, workload and implementation detail.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**For n=6, calculate total iterations of two independent nested n loops.**

**Explain**

Each of the six outer iterations runs six inner iterations.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

**YOU DO — INDEPENDENT PRACTICE**

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

For n=5 and n=10, calculate $n^2$ and $2^n$ values. [6]

### Question 2

Explain time growth of a nested n×n loop and classify. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

A problem has 30 binary choices. Evaluate full subset enumeration and one way to reduce exploration. [6]

**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**An $O(2^n)$ approach grows only twice as much when n doubles.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.